approaches invoke contortions in order to not accept superpositions like the one written above as the true and complete description of nature. As Everett would later put it, "The Copenhagen Interpretation is hopelessly incomplete because of its *a priori* reliance on classical physics . . . as well as a philosophic monstrosity with a 'reality' concept for the macroscopic world and denial of the same for the microcosm."

Everett's prescription was simple: stop contorting yourself. Accept the reality of what the Schrödinger equation predicts. Both parts of the final wave function are actually there. They simply describe separate, never-to-interact-again worlds.

Everett didn't introduce anything new into quantum mechanics; he removed some extraneous clunky pieces from the formalism. Every non-Everettian version of quantum mechanics is, as physicist Ted Bunn has put it, a "disappearing worlds" theory. If the multiple worlds bother you, you have to fiddle with either the nature of quantum states or their ordinary evolution in order to get rid of them. Is it worth it?

There's a looming question here. We're familiar with how wave functions represent superpositions of different possible measurement outcomes. The wave function of an electron can put it in a superposition of various possible locations, as well as in a superposition of spin-up and spin-down. But we were never tempted to say that each part of the superposition was a separate "world." Indeed, it would have been incoherent to do so. An electron that is in a pure spin-up state with respect to the vertical axis is in a superposition of spin-up and spin-down with respect to the horizontal axis. So does that describe one world, or two?

Everett suggested that it is logically consistent to think of superpositions involving macroscopic objects as describing separate worlds. But at the time he was writing, physicists hadn't yet developed the technical tools necessary to turn this into a complete picture. That understanding only came later, with the appreciation of a phenomenon known as *decoherence*. Introduced in 1970 by the German physicist Hans Dieter Zeh, the idea of decoherence has become a central part of how physicists think about quantum dynamics. To the modern Everettian, decoherence is absolutely